

Data Cleaning for Quant Research

Classify data defects, apply conservative repairs, and preserve auditable evidence.

Library of the Quant

Educational reference manual

Manual Profile

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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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How to use this manual

BEGINNER

1. START FROM THE CONSUMER

State the decision, cutoff, grain, and tolerance before labeling a record bad. The same event can be valid for audit, invalid for execution, and useful for incident diagnosis.

2. PRESERVE BEFORE REPAIRING

Retain delivered payloads and canonical events with source clocks and identifiers. Curated values should always trace to originals, rules, evidence, and versions.

3. CLASSIFY THE DEFECT

Separate structural failures, semantic violations, missingness reasons, contextual anomalies, and statistical surprise. Different causes require different actions and confidence.

4. PREFER CONSERVATIVE ACTIONS

Quarantine uncertainty, expire stale states, and publish alternate views rather than silently deleting or smoothing. Repair only when evidence defines a defensible value.

5. VALIDATE THE WHOLE PATH

Test keys, schemas, clocks, units, calendars, actions, distribution shape, and reconciliation. Rebuild from fixtures and verify point-in-time behavior.

6. OPERATE THE CONTRACT

Monitor quality by source, field, instrument, session, and consumer. Connect warnings, degraded modes, hard stops, fallback rules, ownership, and recovery evidence.

DECISION STANDARD

Promote a dataset only when its source snapshot, cleaning policy, quality results, lineage, limitations, and compatible consumers are explicit and reproducible.

Notation and the cleaning record

BEGINNER

X_T AND R_T

Observed field and return at event or interval time t . Identify the clock, instrument, unit, and whether each value is raw, adjusted, or synthetic.

T_EVENT, T_ARRIVAL, T_PUBLISH

Economic event time, collector arrival time, and curated availability time. Keeping them separate supports ordering, latency analysis, and causal eligibility.

K AND G

K is a declared primary key; G is row grain. Uniqueness checks are meaningful only after one row per what has been stated.

M_T AND R_T

Missing indicator and reason code. These distinguish expected absence, outage, invalidation, censoring, and not-yet-available values.

MED, MAD, IQR

Robust center and dispersion statistics. They support anomaly evidence while preserving the distinction between unusual and erroneous.

AGE_T AND W

Observation age and rolling reference window. Both require session and cutoff rules so a carried value never masquerades as fresh.

Q AND S

Quality outcome vector Q and immutable snapshot identifier S. Models should reference the full status and exact data artifact.

CORE INVARIANT

No cleaning decision should make earlier point-in-time outputs depend on later arrivals. Future perturbation tests provide direct evidence.

1. LAND

Write raw payloads immutably with source, delivery, timestamps, checksums, and access controls. Confirm transport completeness before parsing content.

2. NORMALIZE

Resolve schema, identity, units, clocks, calendars, and correction semantics. Invalid records enter quarantine without disappearing.

3. DIAGNOSE

Apply structural invariants, semantic rules, contextual checks, robust anomaly scores, and cross-source evidence in a governed hierarchy.

4. DECIDE

Pass, expire, exclude, repair, impute, quarantine, or escalate under a consumer-specific policy. Preserve original value and every fired reason.

5. VALIDATE

Reconcile counts and economics, compare distributions, run fixtures, test idempotence, and prove point-in-time behavior.

6. PUBLISH

Create an immutable manifest containing inputs, schemas, policies, code, checksums, quality reports, lineage, and approved consumers.

7. MONITOR

Track completeness, timeliness, validity, coverage, drift, revisions, and downstream impact. Connect thresholds to safe actions.

Cleaning starts with fitness for use

BEGINNER

ONE-SENTENCE MEANING

Data cleaning is the controlled process of detecting, explaining, and handling records that are unfit for a declared decision. A value can be valid for audit yet unsuitable for a return feature, execution model, or risk mark.

MEMORY JOGGER

There is no universally clean table. Name the consumer, decision time, grain, tolerance, and economic meaning before writing a rule.

WHY IT MATTERS IN QUANT RESEARCH

A generic delete-outliers pass can erase crisis observations or preserve bad quotes that happen to fall inside a numeric band. Fitness criteria force every action to support a concrete research claim.

FORMULA INTUITION

Define quality as $Q(\text{record}, \text{use_case}, \text{cutoff})$ rather than as a property of the number alone. The same print may pass archival integrity while failing regular-session bar eligibility.

SIMPLE MARKET EXAMPLE

A late auction print belongs in the trade archive but not in a continuous-session midpoint feature. Keeping both views preserves evidence and prevents the feature from inventing a price jump.

PYTHON / SYSTEM IMPLEMENTATION

Create a versioned cleaning contract with required fields, admissible states, source clocks, repair actions, and stop conditions. Emit pass, fail, or quarantine reason codes rather than one opaque clean flag.

CONFUSIONS TO AVOID

Clean does not mean smooth, complete, or normally distributed. Markets contain real jumps, sparse trading, stale states, and structural breaks that a cosmetic rule can destroy.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dataset registry binds each model and backtest to a use-case-specific quality policy. Raw evidence remains available so a later policy can reproduce or challenge every decision.

RELATED CONCEPTS AND QUICK RECALL

Related: data contract, eligibility, semantic validity, audit trail. Recall task: Choose one price record and explain why it can be valid for storage but invalid for a trading feature.

Immutable raw data and curated views

BEGINNER

ONE-SENTENCE MEANING

An immutable raw layer preserves delivered payloads, while curated layers normalize, classify, repair, and publish records under explicit policies. The raw layer is evidence; a curated layer is an interpretation that can be rebuilt.

MEMORY JOGGER

Never make the only copy look nicer. Preserve what arrived, then create named views that explain every transformation.

WHY IT MATTERS IN QUANT RESEARCH

Vendor corrections, parser bugs, and changing research needs require replay from original bytes or canonical events. Destructive edits remove the ability to diagnose whether a surprising backtest came from markets, vendors, or code.

FORMULA INTUITION

$\text{Curated_t} = F(\text{raw_snapshot}, \text{schema_version}, \text{policy_version}, \text{code_hash})$. Reproducibility requires every argument to F , not only the final rows.

SIMPLE MARKET EXAMPLE

A vendor later corrects a decimal-place error. An append-only correction plus a new curated snapshot shows which experiments used the original value and which used the repair.

PYTHON / SYSTEM IMPLEMENTATION

Write raw objects once with checksums and source metadata; normalize into append-only tables; publish immutable manifests for research. Keep repair links from each curated row to source event IDs.

CONFUSIONS TO AVOID

Raw does not mean trustworthy or ready for analysis, and curated does not mean true. Both layers need schemas, retention, access controls, and observable failure modes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Ingestion owns delivered evidence, normalization owns canonical semantics, and dataset builders own use-specific eligibility. Research artifacts reference the exact manifest rather than a mutable path.

RELATED CONCEPTS AND QUICK RECALL

Related: bronze-silver-gold layers, snapshot, checksum, lineage. Recall task: List the identifiers required to rebuild one curated bar from immutable raw events.

Grain, keys, and schema integrity

BEGINNER

ONE-SENTENCE MEANING

Dataset grain states what one row represents, while keys and schema define its identity, types, units, and required relationships. Cleaning cannot be correct when rows mix trades, bars, instruments, or observation clocks.

MEMORY JOGGER

Say one row per what before checking nulls or ranges. A technically valid column can still be wrong because it belongs to another grain.

WHY IT MATTERS IN QUANT RESEARCH

Mixed grain creates duplicate joins, inflated aggregates, and misleading missingness. Structural validation catches these failures before statistical filters hide their symptoms.

FORMULA INTUITION

For key K , require $\text{count}(\text{rows}) = \text{count}(\text{distinct } K)$ when uniqueness is promised. For a declared many-to-one join, verify every left key matches at most one right validity interval.

SIMPLE MARKET EXAMPLE

Joining daily fundamentals to intraday quotes without a temporal key repeats one fact thousands of times. A later sum appears plausible in code but has no economic interpretation.

PYTHON / SYSTEM IMPLEMENTATION

Validate column types, nullability, allowed enums, key uniqueness, referential integrity, sort expectations, and join cardinality at every boundary. Store the schema fingerprint with the output manifest.

CONFUSIONS TO AVOID

A surrogate row ID does not repair unclear economic identity. Automatic type coercion can turn failed timestamps into nulls and silently shift a structural error into a missing-data problem.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The contract registry blocks incompatible producers and consumers before data reaches feature engineering. Failed batches enter quarantine with exact key and schema evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: primary key, cardinality, schema evolution, referential integrity. Recall task: Define the grain and primary key for a one-minute instrument bar, then name two invalid duplicates.

Knowledge time and point-in-time eligibility

BEGINNER

ONE-SENTENCE MEANING

Point-in-time cleaning preserves when a record became knowable, when it was economically effective, and which later corrections existed at each cutoff. A historically accurate value can still leak if its corrected version was unavailable then.

MEMORY JOGGER

Clean by the clock the strategy could observe, not by the timestamp most convenient for today. Availability and event time answer different questions.

WHY IT MATTERS IN QUANT RESEARCH

Backtests can gain artificial accuracy from revised fundamentals, late trade corrections, or backfilled reference data. Preserving knowledge time makes historical features causal and replayable.

FORMULA INTUITION

A record is eligible at cutoff t only if $\text{available_time} \leq t$, $\text{effective_from} \leq t < \text{effective_to}$, and its revision was not superseded before t . Tolerances and session rules add further constraints.

SIMPLE MARKET EXAMPLE

An earnings value is restated two months later. The current database should show the restatement for reporting, while an as-of training snapshot must retain the originally published value until the correction arrived.

PYTHON / SYSTEM IMPLEMENTATION

Store event, arrival, normalized, published, and revision timestamps separately. Build as-of joins that return matched time, age, revision ID, and rejection reason.

CONFUSIONS TO AVOID

Sorting by event time alone can move a late record backward into a feature. Backfilling a corrected value across all history is convenient reporting but invalid model training.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The snapshot service materializes data by decision cutoff and vendor vintage. Validation perturbs future records to confirm that earlier features do not change.

RELATED CONCEPTS AND QUICK RECALL

Related: bitemporal data, as-of join, revision, look-ahead bias. Recall task: Write the eligibility test for a corrected fundamental used by a decision made before the correction arrived.

Units, conventions, and normalization

BEGINNER

ONE-SENTENCE MEANING

Normalization converts source-specific units, signs, currencies, scales, and field conventions into a canonical representation while retaining the original values. A number without units and convention is incomplete data.

MEMORY JOGGER

Most dangerous cleaning errors are semantic, not numeric. Software can sum shares, lots, contracts, basis points, and decimals without raising an exception.

WHY IT MATTERS IN QUANT RESEARCH

Return, notional, liquidity, and cost estimates depend on contract multipliers, FX direction, price scale, and corporate-action state. Unit drift can produce stable-looking but economically absurd features.

FORMULA INTUITION

Notional = price * quantity * contract_multiplier; converted_notional = notional * FX_rate under a declared base/quote direction and timestamp. Dimensional checks should precede aggregation.

SIMPLE MARKET EXAMPLE

A futures feed switches size from contracts to underlying units while values remain positive integers. Median checks may pass, but estimated liquidity jumps by the multiplier.

PYTHON / SYSTEM IMPLEMENTATION

Attach unit metadata to the schema, use integer ticks where appropriate, and version every conversion table. Reconcile canonical values back to source statements and preserve both columns.

CONFUSIONS TO AVOID

Standardization to z-scores does not resolve incorrect units; it can conceal them. Percent, decimal return, basis points, and price points are distinct conventions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

A unit service supplies multipliers, currencies, tick sizes, and FX conversions to ingestion, features, costs, and risk. Contract changes create new semantic versions rather than silent rewrites.

RELATED CONCEPTS AND QUICK RECALL

Related: dimensional analysis, contract multiplier, FX convention, tick size. Recall task: Name the metadata needed before multiplying a futures price by its reported size.

Calendars, sessions, and time zones

BEGINNER

ONE-SENTENCE MEANING

Calendar cleaning maps timestamps to valid trading sessions, auctions, holidays, early closes, and timezone transitions. It distinguishes a genuinely missing observation from a period when no observation should exist.

MEMORY JOGGER

A regular grid is only meaningful after the market calendar is known. Midnight UTC is not a universal trading-day boundary.

WHY IT MATTERS IN QUANT RESEARCH

Missing-bar rates, overnight returns, feature windows, and cross-market joins change when sessions are mislabeled. Daylight-saving shifts can alter market overlap without changing local opening times.

FORMULA INTUITION

Expected observations equal the sum over declared session intervals of eligible grid points. Missingness is observed minus expected, not observed minus every wall-clock interval.

SIMPLE MARKET EXAMPLE

An early-close day legitimately contains fewer U.S. equity bars. Treating the afternoon as an outage creates false alerts and forward-filled synthetic liquidity.

PYTHON / SYSTEM IMPLEMENTATION

Use a versioned exchange calendar and retain UTC time plus local session label. Test holidays, half-days, auctions, overnight futures, and daylight transitions with fixed fixtures.

CONFUSIONS TO AVOID

Forward-filling across a session boundary makes yesterday's close look current. Calendar date and trading session label can differ for instruments whose session spans midnight.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The shared calendar service controls ingestion expectations, bar construction, feature windows, labels, and execution simulation. One definition prevents layer-specific time drift.

RELATED CONCEPTS AND QUICK RECALL

Related: trading calendar, session label, timezone, early close. Recall task: Explain why an absent 15:30 bar is an error on one day and expected on another.

Provenance and record-level lineage

BEGINNER

ONE-SENTENCE MEANING

Provenance records where data came from, while lineage records how each output was derived through versions, parameters, and source rows. Together they turn cleaning from an invisible edit into an auditable computation.

MEMORY JOGGER

Every changed value needs an answer to who, what, when, why, and from which evidence. Aggregate metadata alone is insufficient for disputed records.

WHY IT MATTERS IN QUANT RESEARCH

When performance changes after a refresh, lineage separates vendor revisions from policy changes and code regressions. It also supports targeted rollback without discarding unrelated clean data.

FORMULA INTUITION

A lineage edge can be modeled `output_id -> {source_ids, transform_id, policy_id, run_id}`. A snapshot hash addresses the complete ordered set of files and metadata.

SIMPLE MARKET EXAMPLE

A repaired bar high traces to three eligible trades and excludes one canceled print. An investigator can reproduce both the included path and the reason-coded exclusion.

PYTHON / SYSTEM IMPLEMENTATION

Emit source IDs, rule outcomes, old and new values, transform code hash, run ID, and snapshot ID. Store compact row-level links or deterministic recipes when full expansion is too large.

CONFUSIONS TO AVOID

A notebook filename is not lineage, and a latest table is not a reproducible source. Logging only failed records hides how passing records were interpreted.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The catalog exposes lineage to feature, model, backtest, and production-decision artifacts. Incident tooling can query every consumer affected by a defective source range.

RELATED CONCEPTS AND QUICK RECALL

Related: provenance, audit log, manifest, data observability. Recall task: Trace one model input backward through feature, curated table, normalized event, and raw payload.

Deterministic and idempotent cleaning

BEGINNER

ONE-SENTENCE MEANING

A deterministic cleaning job produces the same output from the same ordered inputs and versions, while idempotence makes safe retries produce no duplicate effects. These properties are essential when feeds replay or jobs restart.

MEMORY JOGGER

Retries are normal operating behavior. A clean pipeline must treat the second delivery as evidence already processed, not as new market activity.

WHY IT MATTERS IN QUANT RESEARCH

Non-deterministic ordering changes opens, closes, deduplication winners, and as-of matches. Non-idempotent writes inflate volume and break confidence in recovery.

FORMULA INTUITION

Idempotence requires $f(f(S,e),e)=f(S,e)$ for repeated event e . Determinism also requires stable sorting, explicit tie-breaks, fixed dependencies, and controlled randomness.

SIMPLE MARKET EXAMPLE

Two trade messages share a timestamp but different sequence numbers. Sorting only by time can change the bar close across runs; ordering by venue sequence makes the result stable.

PYTHON / SYSTEM IMPLEMENTATION

Use event identities, transactional checkpoints, deterministic total order, fixed policy versions, and atomic publication. Compare output hashes on controlled replay.

CONFUSIONS TO AVOID

Dropping duplicate rows after concatenation is not enough if the chosen winner depends on file order. Current-time defaults and unseeded stochastic imputers also break deterministic rebuilds.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The orchestration layer records offsets, input manifests, code hashes, and output checksums. Recovery tests replay known windows and require byte-stable or semantically stable results.

RELATED CONCEPTS AND QUICK RECALL

Related: idempotence, total order, checkpoint, reproducibility. Recall task: Describe how a retry can double volume even when every individual trade row looks valid.

Missingness is a taxonomy, not one null

BEGINNER

ONE-SENTENCE MEANING

Missing data can mean not expected, not observed, not delivered, invalidated, censored, unavailable yet, or intentionally suppressed. Converting every case to the same null destroys information needed for correct repair.

MEMORY JOGGER

Ask why the value is absent before asking how to fill it. The reason often predicts both bias and the safest downstream action.

WHY IT MATTERS IN QUANT RESEARCH

Market inactivity, feed outages, delistings, and eligibility filters create different selection mechanisms. Models trained on silently filled values can learn data availability rather than price behavior.

FORMULA INTUITION

Track indicator M_t and reason R_t alongside value X_t ; estimate $P(M=1 \mid \text{instrument, time, source, state})$. Missing completely at random is rarely a safe default in markets.

SIMPLE MARKET EXAMPLE

No quote during a halt is structurally different from a lost packet during active trading. Both create an empty cell, but only one should trigger feed recovery.

PYTHON / SYSTEM IMPLEMENTATION

Create reason-coded nulls, expected-grid tables, observation age, and coverage denominators. Choose exclusion, explicit missing feature, model-based imputation, or escalation by use case.

CONFUSIONS TO AVOID

Zero is usually an observed value, not a missing marker. Forward-fill can turn an outage into apparent price stability and overstate available liquidity.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The quality layer publishes value, status, reason, age, and expectedness. Feature builders consume that bundle and prevent imputation across forbidden boundaries.

RELATED CONCEPTS AND QUICK RECALL

Related: missing-data mechanism, censoring, coverage, staleness. Recall task: Classify an absent quote during a halt and an absent quote during an active session, then choose different actions.

Duplicate delivery and event identity

BEGINNER

ONE-SENTENCE MEANING

A duplicate occurs when the same economic event is delivered more than once, not merely when two rows share equal values. Reliable deduplication requires stable identity, correction semantics, and a declared retention policy.

MEMORY JOGGER

Equal rows may be separate events; unequal rows may be revisions of one event. Deduplicate by meaning, not by visual similarity.

WHY IT MATTERS IN QUANT RESEARCH

Duplicates inflate volume, counts, volatility estimators, and labels, while aggressive row-level dropping can erase legitimate repeated trades. The error is especially damaging when retries correlate with volatile periods.

FORMULA INTUITION

Candidate key may include source, venue, channel, session, sequence, event_type, and event_id. Compare duplicate rate by partition and delivery attempt, not only globally.

SIMPLE MARKET EXAMPLE

Two identical 100-share trades at the same price can be legitimate consecutive executions. A replayed message with the same sequence and event ID is one event delivered twice.

PYTHON / SYSTEM IMPLEMENTATION

Persist source identity, delivery ID, and payload hash; apply idempotent upserts or append with canonical winner rules. Link cancels and corrections rather than deleting history.

CONFUSIONS TO AVOID

Dropping identical rows over all columns is neither a business rule nor a correction model. Timestamp alone is weak identity because venues emit multiple events at the same instant.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Ingestion owns transport deduplication, normalization resolves revisions, and curated datasets select active versions. Metrics expose replay rate and winner reasons by source.

RELATED CONCEPTS AND QUICK RECALL

Related: event identity, replay, correction, canonical record. Recall task: Explain why two equal trade rows may both be valid while two unequal rows may represent one corrected event.

Bad ticks and impossible prices

BEGINNER

ONE-SENTENCE MEANING

A bad tick is a price or size observation inconsistent with market rules, neighboring evidence, or source semantics. Detection should combine hard invariants with contextual checks rather than a single return threshold.

MEMORY JOGGER

A large move is not automatically bad, and a small move can still violate the tick grid. Use economic constraints before statistical surprise.

WHY IT MATTERS IN QUANT RESEARCH

One decimal error can dominate returns, volatility, stops, and model labels across many windows. Deleting real jumps, however, removes exactly the tail behavior risk systems must learn.

FORMULA INTUITION

Check $\text{price} > 0$, tick-grid distance near zero, venue bands, spread geometry, cross-source consensus, and robust return score. A robust score can use $|r_t - \text{median}(r)| / (1.4826 * \text{MAD})$.

SIMPLE MARKET EXAMPLE

A print at 10.03 is valid when tick size is 0.01, while 10.035 violates the grid. A 12% earnings gap may be statistically extreme but confirmed across quotes and venues.

PYTHON / SYSTEM IMPLEMENTATION

Stage rules from impossible to suspicious; preserve the record and evidence; quarantine uncertain cases. Recompute derived bars only after reason-coded review or corroboration.

CONFUSIONS TO AVOID

Clipping every return beyond five standard deviations manufactures thin tails. Comparing only with the previous trade also mistakes bid-ask bounce or auction transitions for errors.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The event-quality service emits hard-fail, soft-alert, corroborated-jump, and unresolved states. Bar and label builders consume policy-specific eligibility without mutating raw trades.

RELATED CONCEPTS AND QUICK RECALL

Related: price bands, tick size, robust return, cross-source validation. Recall task: Design evidence that distinguishes a decimal error from a genuine news gap.

Outliers versus rare but real events

BEGINNER

ONE-SENTENCE MEANING

An outlier is an observation unusual relative to a reference distribution, while an error is a failure of measurement or semantics. The two sets overlap but neither contains the other.

MEMORY JOGGER

Statistical rarity asks how surprising; data quality asks whether it happened and means what the schema claims. Do not substitute one question for the other.

WHY IT MATTERS IN QUANT RESEARCH

Quant research depends on tails for risk, capacity, and stress behavior. Removing rare real events creates optimistic drawdowns, unstable thresholds, and models that fail precisely when protection matters.

FORMULA INTUITION

Estimate robust distance within a relevant regime and compare with independent evidence. Flag probability or score should inform review, not automatically define truth.

SIMPLE MARKET EXAMPLE

A flash crash print confirmed across venues is a real tail event. A single unconfirmed zero price from one vendor is a likely defect even if a broad historical rule cannot classify it.

PYTHON / SYSTEM IMPLEMENTATION

Store anomaly score, rule evidence, corroborating sources, regime label, and final disposition separately. Maintain analysis views both with and without unresolved points.

CONFUSIONS TO AVOID

A Gaussian sigma rule is fragile when returns are heavy-tailed and volatility changes. Winsorizing labels before validation can hide the economic cost of extreme outcomes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Exploration compares raw and governed views; risk validation requires preserved genuine extremes. Production escalates unresolved high-impact anomalies instead of silently smoothing them.

RELATED CONCEPTS AND QUICK RECALL

Related: tail event, anomaly, measurement error, robust statistics. Recall task: Give one observation that is extreme but valid and one that is moderate but semantically invalid.

Stale quotes and observation age

BEGINNER

ONE-SENTENCE MEANING

A stale value remains unchanged or unrefreshed beyond a context-dependent age, even though it may still be syntactically valid. Staleness depends on instrument activity, session state, venue, and intended action horizon.

MEMORY JOGGER

Last observed does not mean currently tradable. Carry age and market state with every forward-carried value.

WHY IT MATTERS IN QUANT RESEARCH

Stale midpoints understate volatility, distort spreads, create false cross-asset lead-lag, and overstate executable liquidity. Illiquid instruments require different thresholds from active futures.

FORMULA INTUITION

$\text{age_t} = \text{decision_time} - \text{last_update_time}$; compare age with empirical inter-arrival quantiles conditional on instrument and session. Unchanged-value runs are supporting evidence, not a universal rule.

SIMPLE MARKET EXAMPLE

A quote unchanged for three seconds may be stale in an active index future but normal in a thin bond. Venue-wide silence during an active session suggests a feed issue.

PYTHON / SYSTEM IMPLEMENTATION

Compute observation age, update intensity, sequence health, venue coverage, and last executable state. Expire or downweight records through policy rather than overwriting their timestamps.

CONFUSIONS TO AVOID

Forward-filling changes availability representation but creates no new observation. A constant fundamental value can remain current for months, so unchanged-run logic is field-specific.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Features include age and freshness state; execution refuses expired quotes; monitoring aggregates stale share by source and venue. Fallbacks must be explicitly compatible with the consumer.

RELATED CONCEPTS AND QUICK RECALL

Related: freshness, heartbeat, quote fade, forward fill. Recall task: Set different freshness logic for a liquid future and an illiquid bond, and explain the conditioning variables.

Crossed and locked markets

BEGINNER

ONE-SENTENCE MEANING

A locked quote has bid equal to ask, while a crossed quote has bid above ask. These states can be real and transient, or symptoms of asynchronous feeds, stale venues, eligibility differences, or reconstruction errors.

MEMORY JOGGER

Impossible-looking spread geometry is a diagnostic clue, not an automatic delete command. Identify whether the quotes were jointly observable and executable.

WHY IT MATTERS IN QUANT RESEARCH

Negative spreads break midpoint, cost, and execution logic, yet blanket removal can bias periods of fast markets. The duration and source composition often reveal the cause.

FORMULA INTUITION

$\text{spread} = \text{ask} - \text{bid}$; evaluate sign, duration, venue timestamps, sequence continuity, and observation age. Consolidated crosses should be reconstructed from synchronized eligible venue states.

SIMPLE MARKET EXAMPLE

A brief lock around an auction transition may be legitimate. A persistent cross caused by one stale ask and a fresh bid is a data-quality failure with clear provenance.

PYTHON / SYSTEM IMPLEMENTATION

Preserve venue quotes, rebuild consolidated state under one eligibility policy, and classify lock or cross episodes by duration and evidence. Exclude only from consumers that require a positive executable spread.

CONFUSIONS TO AVOID

Replacing negative spreads with zero hides both the defect and its effect on simulated costs. Midpoint computed from asynchronous legs may never have existed in the market.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Quote-state health feeds feature quality and execution safeguards. Monitoring tracks locked and crossed duration by venue, session phase, and reconstruction version.

RELATED CONCEPTS AND QUICK RECALL

Related: NBBO, spread, synchronization, venue eligibility. Recall task: List the evidence needed to decide whether a crossed market is real, transient, or caused by stale state.

Sequence gaps and out-of-order events

BEGINNER

ONE-SENTENCE MEANING

Event streams can arrive late, out of order, duplicated, or with missing sequence ranges. Correct reconstruction requires a source-specific ordering contract and a bounded policy for disorder.

MEMORY JOGGER

Arrival order is not always market order, but future sorting is not always causally available. Preserve both sequence and delivery clocks.

WHY IT MATTERS IN QUANT RESEARCH

A single missing book update can corrupt every later depth snapshot until recovery. Silent reordering also changes bar opens, closes, and event-driven features.

FORMULA INTUITION

For monotone sequence s_i , require $s_i = s_{(i-1)} + 1$ unless the protocol declares a reset or channel split. Watermarks define how long to wait before finalizing event-time windows.

SIMPLE MARKET EXAMPLE

An add message arrives after its cancel because network paths differ. Applying arrival order leaves a phantom order; sequence order restores state if both messages are available before the decision cutoff.

PYTHON / SYSTEM IMPLEMENTATION

Detect gaps, buffer bounded disorder, checkpoint verified state, and request replay or snapshot recovery. Record watermark, late-event action, and whether historical outputs were revised.

CONFUSIONS TO AVOID

Sorting the complete day after close produces accurate history but can leak future availability into an intraday simulation. Sequence resets at session or reconnect are not duplicates.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The stream processor exposes completeness and watermark state to consumers. Derived outputs remain provisional until policy finalization, then revisions create new snapshots.

RELATED CONCEPTS AND QUICK RECALL

Related: sequence number, watermark, late data, state reconstruction. Recall task: Explain how a complete end-of-day sort can still make an intraday backtest unrealistic.

Corporate actions and artificial jumps

BEGINNER

ONE-SENTENCE MEANING

Splits, dividends, mergers, spinoffs, and symbol changes alter the relationship between raw prices, shares, and economic returns. Cleaning must distinguish accounting discontinuities from market moves without erasing original observations.

MEMORY JOGGER

An adjustment is a versioned economic transformation, not a visual patch. Preserve raw, split-adjusted, and total-return views separately.

WHY IT MATTERS IN QUANT RESEARCH

Unapplied actions create extreme returns, false volatility, broken stops, and inconsistent market capitalization. Future-known adjustment factors can also leak corporate events backward into training data.

FORMULA INTUITION

For split factor A_t , a comparable price may use $P_{adj,t} = P_{raw,t} / A_t$ under a stated factor chain. Total return additionally handles cash distributions and reinvestment convention.

SIMPLE MARKET EXAMPLE

A two-for-one split halves raw price and doubles shares without a 50% economic loss. A cash dividend affects price return and total return differently.

PYTHON / SYSTEM IMPLEMENTATION

Store announcement, ex-date, effective time, payment date, terms, revision, and source. Generate point-in-time factor chains and reconcile price, shares, and market value across the event.

CONFUSIONS TO AVOID

Back-adjusted latest history can change after vendor revisions. Treating every large close-to-close move as a corporate action can hide genuine crashes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The action service publishes governed factor views to bars, returns, universes, and fundamentals. Dataset manifests record both raw source and adjustment policy ID.

RELATED CONCEPTS AND QUICK RECALL

Related: split factor, total return, ex-date, factor chain. Recall task: Describe tests that distinguish an unadjusted split from a real price collapse.

Instrument identity and symbol changes

BEGINNER

ONE-SENTENCE MEANING

Symbols are changing labels, while instruments and issuers require stable internal identity with effective-dated mappings. Cleaning joins must resolve the correct security at the record's economic time.

MEMORY JOGGER

Ticker equality is not identity. Reuse, mergers, share classes, rolls, and venue changes make text joins unsafe.

WHY IT MATTERS IN QUANT RESEARCH

Identity errors splice unrelated histories, create impossible returns, misapply actions, and introduce survivorship. These failures often pass numeric checks because each source value is individually valid.

FORMULA INTUITION

A mapping is eligible when $\text{effective_from} \leq \text{event_time} < \text{effective_to}$ and source namespace matches. Validity intervals for one source key must not overlap unless ambiguity is explicitly modeled.

SIMPLE MARKET EXAMPLE

A delisted ticker is later reused by another company. A timeless join merges two securities into one apparently long history with a giant discontinuity.

PYTHON / SYSTEM IMPLEMENTATION

Use surrogate instrument IDs and effective-dated vendor, symbol, venue, issuer, currency, and contract mappings. Quarantine unresolved or multiply matched records.

CONFUSIONS TO AVOID

Permanent identifiers do not always imply economic continuity; contract expiry or ADR ratio changes may require a new analysis identity. Backfilled current symbols can conceal historical transitions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The security master resolves every normalized event and publishes mapping health. Universe construction and corporate-action processing share the same point-in-time identity service.

RELATED CONCEPTS AND QUICK RECALL

Related: security master, symbology, effective dating, survivorship. Recall task: Construct a safe temporal join for a ticker that changes after a merger.

Bar integrity and aggregation anomalies

BEGINNER

ONE-SENTENCE MEANING

OHLCV bars are policy-defined summaries of eligible events, so cleaning must validate both numeric relationships and the event path that produced them. A syntactically valid bar can still be built from the wrong session or trade set.

MEMORY JOGGER

Check the summary and its recipe. Open, high, low, close, and volume have meaning only after boundaries and eligibility are fixed.

WHY IT MATTERS IN QUANT RESEARCH

Bad bars contaminate technical features, volatility, targets, and simulated fills across many downstream windows. Rebuilding from events is safer than patching isolated fields.

FORMULA INTUITION

Require $\text{low} \leq \min(\text{open}, \text{close}) \leq \max(\text{open}, \text{close}) \leq \text{high}$ and $\text{volume} \geq 0$. Also reconcile count, notional, and first or last eligible event under the bar policy.

SIMPLE MARKET EXAMPLE

A bar with high below close is structurally impossible. A bar with plausible OHLC can still include a late block or cross an early-close boundary.

PYTHON / SYSTEM IMPLEMENTATION

Validate interval closure, session membership, eligible trade conditions, event ordering, empty-bar policy, and late revisions. Retain bar-to-event lineage and deterministic rebuild tests.

CONFUSIONS TO AVOID

Replacing an invalid high with $\max(\text{open}, \text{close})$ creates an internally consistent fiction. Forward-filled empty bars need zero observed volume and explicit synthetic status.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The bar service publishes policy-versioned raw and adjusted products. Feature builders reject bars whose finalization, lineage, or quality status does not satisfy the model contract.

RELATED CONCEPTS AND QUICK RECALL

Related: OHLCV, resampling, sale conditions, empty bar. Recall task: Name one structural bar invariant and one path-level check that numeric fields alone cannot prove.

Vendor revisions and mutable history

BEGINNER

ONE-SENTENCE MEANING

Vendors may correct, backfill, restate, or reprocess historical records after initial delivery. Cleaning must preserve vintages so research can distinguish improved truth from what was actually known.

MEMORY JOGGER

Latest is a useful view, not an immutable source. Every refresh should answer what changed, why, and which artifacts are affected.

WHY IT MATTERS IN QUANT RESEARCH

Mutable inputs can change backtest results without code changes and leak later knowledge into earlier features. Untracked revisions make model comparison and incident analysis unreliable.

FORMULA INTUITION

Represent each logical record with valid time, available time, revision sequence, and supersession link. Compare snapshot hashes and field-level deltas between vintages.

SIMPLE MARKET EXAMPLE

A vendor reprocesses years of sentiment with a new classifier. The latest dump looks historically stable even though those scores were never available to the original strategy.

PYTHON / SYSTEM IMPLEMENTATION

Land each delivery immutably, compute change manifests, classify correction versus methodology change, and rebuild only affected partitions. Require approval for material historical restatements.

CONFUSIONS TO AVOID

Overwriting a partition destroys the old research state. Treating every revision as an error also prevents legitimate data improvement and current reporting accuracy.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The catalog tracks vintage lineage from source to models and backtests. Impact analysis lists affected features, experiments, reports, and live controls before republishing.

RELATED CONCEPTS AND QUICK RECALL

Related: vintage, restatement, change data capture, impact analysis. Recall task: Explain how a methodology reprocess differs from a correction and how each should affect historical experiments.

Robust center and scale

BEGINNER

ONE-SENTENCE MEANING

Robust statistics estimate location and dispersion with limited sensitivity to a small number of extreme observations. Median, interquartile range, and median absolute deviation often reveal structure that mean and standard deviation obscure.

MEMORY JOGGER

Use robust summaries to detect unusual values, not to declare them false. The estimator is a lens, not a truth machine.

WHY IT MATTERS IN QUANT RESEARCH

Market distributions are heavy-tailed, regime-dependent, and contaminated by occasional feed errors. Robust baselines prevent one defect from inflating the threshold used to detect itself.

FORMULA INTUITION

$MAD = \text{median}(|x_i - \text{median}(x)|)$; a normal-consistent scale is approximately $1.4826 \cdot MAD$. $IQR = Q_{0.75} - Q_{0.25}$ gives another resistant spread measure.

SIMPLE MARKET EXAMPLE

One price error multiplies the sample standard deviation and makes a three-sigma rule permissive. Median and MAD remain anchored near the unaffected observations.

PYTHON / SYSTEM IMPLEMENTATION

Compute robust summaries within sensible instrument, field, session, and regime groups. Store sample size, zero-MAD handling, window boundaries, and estimator version.

CONFUSIONS TO AVOID

Robust does not mean immune to clustered contamination or regime shifts. A median can be stale or economically irrelevant when the market reprices rapidly.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Exploration compares classical and robust summaries; cleaning rules use them as soft evidence. Validation measures how thresholds behave during stress and sparse samples.

RELATED CONCEPTS AND QUICK RECALL

Related: median, MAD, IQR, influence function. Recall task: Show how one extreme value affects mean and standard deviation more than median and MAD.

Rolling z-scores and local context

BEGINNER

ONE-SENTENCE MEANING

A rolling z-score compares an observation with a recent local center and scale, making anomaly detection responsive to changing market levels. Its validity depends on window design, estimator robustness, and causal timing.

MEMORY JOGGER

Standardize against the past, never against a window that includes future observations. Local context helps only when the window represents the current mechanism.

WHY IT MATTERS IN QUANT RESEARCH

Global thresholds fail when price levels, volatility, or liquidity change. Rolling scores can separate an unusual move from ordinary high-volatility behavior while supporting online monitoring.

FORMULA INTUITION

$z_t = (x_t - \mu_{(t-1, W)}) / \sigma_{(t-1, W)}$, where the baseline ends before t . Robust variants replace mean and standard deviation with median and scaled MAD.

SIMPLE MARKET EXAMPLE

A 1% return is extreme in a quiet regime but routine during a crisis. A lagged rolling score changes interpretation without using information from later returns.

PYTHON / SYSTEM IMPLEMENTATION

Shift rolling statistics before scoring, require minimum history, and record window, weighting, grouping, and reset boundaries. Compare standard and robust variants.

CONFUSIONS TO AVOID

Overlapping windows create serially dependent scores and can smear one error across many baselines. A long window adapts slowly; a short window can normalize a sustained defect.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature and monitoring libraries share one causal rolling implementation. Alerts retain raw value, baseline, scale, score, and state so reviewers can interpret context.

RELATED CONCEPTS AND QUICK RECALL

Related: rolling window, exponential weighting, standardization, leakage. Recall task: Write a causal rolling-score calculation and explain why an unshifted window leaks.

Hampel filters and local anomaly flags

BEGINNER

ONE-SENTENCE MEANING

A Hampel filter flags observations far from a local median by a multiple of local MAD. It is useful for sparse point contamination when the underlying signal is locally stable and genuine jumps are separately protected.

MEMORY JOGGER

Flag first, replace only under a governed rule. The same local deviation can be an error in one field and valuable information in another.

WHY IT MATTERS IN QUANT RESEARCH

The method resists isolated spikes better than moving-mean filters, but finance contains real discontinuities. Combining it with cross-source and market-state evidence reduces false deletion.

FORMULA INTUITION

Flag x_t when $|x_t - \text{med}_W| > k * 1.4826 * \text{MAD}_W$, subject to minimum observations and zero-scale policy. Use a past-only or operationally available window.

SIMPLE MARKET EXAMPLE

A one-tick sensor spike in a stable reference rate is flagged cleanly. A broad earnings gap confirmed across venues should override the local flag and remain as a valid jump.

PYTHON / SYSTEM IMPLEMENTATION

Return an anomaly flag, score, threshold, window, and corroboration status. If replacement is allowed, preserve original value and use a reason-specific method.

CONFUSIONS TO AVOID

A centered window uses future values and can make historical cleaning noncausal. Consecutive defects can shift the local median enough to evade the filter.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The anomaly service supplies candidate flags to policy and review layers. No consumer receives a silently replaced value without provenance and quality status.

RELATED CONCEPTS AND QUICK RECALL

Related: median filter, MAD, local anomaly, corroboration. Recall task: State when a Hampel flag should lead to quarantine and when it should be overridden as a real market event.

Winsorization and clipping

BEGINNER

ONE-SENTENCE MEANING

Winsorization replaces values beyond chosen bounds with the bounds, while clipping enforces hard limits. These transforms change the empirical distribution and should be applied only for a declared modeling reason, never to disguise poor source quality.

MEMORY JOGGER

Capping limits influence; it does not prove an observation was wrong. Keep raw values and measure the decision impact of every cap.

WHY IT MATTERS IN QUANT RESEARCH

Some estimators are dominated by extremes, but trading risk often lives in those same extremes. Careful use may stabilize a feature while uncapped targets and stress metrics preserve economic tails.

FORMULA INTUITION

$x'_i = \min(\max(x_i, L), U)$, where L and U may be fixed economic bounds or training-only quantiles. Bounds must be learned inside each training fold.

SIMPLE MARKET EXAMPLE

Capping a noisy ratio feature at training percentiles may improve stability. Capping realized losses before evaluating drawdown produces misleading risk.

PYTHON / SYSTEM IMPLEMENTATION

Separate source-error quarantine from model transform; fit bounds on training data; store cap indicator and distance; evaluate capped share by regime and asset.

CONFUSIONS TO AVOID

Global quantile caps mix instruments and time periods with different scales. Applying bounds before the train-test split leaks distribution information from the future.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature registry records raw input, cap policy, fit sample, and transformed output. Risk reports retain uncapped outcomes and show sensitivity to the transform.

RELATED CONCEPTS AND QUICK RECALL

Related: winsorization, clipping, robust loss, influence. Recall task: Choose a case where feature capping is defensible and a case where it would invalidate risk measurement.

Interpolation and forward filling

BEGINNER

ONE-SENTENCE MEANING

Interpolation estimates values between observations, while forward filling carries the last observation across a grid. Both create derived values rather than new market evidence and require strict boundaries, age, and method labels.

MEMORY JOGGER

A filled cell is synthetic. Preserve observed status and never infer liquidity, volume, or tradability from the existence of a filled price.

WHY IT MATTERS IN QUANT RESEARCH

Regular grids can simplify models, but improper filling understates volatility, creates lead-lag relationships, and crosses sessions or halts. The method must match the field's economic behavior.

FORMULA INTUITION

Linear interpolation uses $x(t) = x_a + (x_b - x_a)(t - t_a) / (t_b - t_a)$, which uses future endpoint b. Forward fill $x(t) = x_a$ is causal but grows stale.

SIMPLE MARKET EXAMPLE

Interpolating a missing midpoint for visualization may be acceptable after the fact. Using that same interpolation in an intraday signal gives the strategy a future quote.

PYTHON / SYSTEM IMPLEMENTATION

Set maximum gap, session boundaries, field eligibility, and causal mode. Emit imputed flag, method, source endpoints, observation age, and uncertainty penalty.

CONFUSIONS TO AVOID

Prices, sizes, returns, and categorical states require different rules. Forward-filling volume creates trades that never happened; interpolating through a halt invents a path.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dataset builder offers governed fill operators with explicit offline and online semantics. Model features can include age and missing indicators or reject synthetic values entirely.

RELATED CONCEPTS AND QUICK RECALL

Related: imputation, regular grid, staleness, causal transform. Recall task: Explain why linear interpolation is usually unsuitable for a real-time feature even when it looks smooth on a chart.

Resampling and synchronization

BEGINNER

ONE-SENTENCE MEANING

Resampling maps irregular events to declared intervals, while synchronization aligns instruments or sources on a common information clock. Correct cleaning requires explicit boundaries, finalization, and maximum-age rules.

MEMORY JOGGER

A row-aligned matrix is not automatically time-aligned information. Every cell needs a cutoff and an age.

WHY IT MATTERS IN QUANT RESEARCH

Cross-asset covariance, spread signals, and portfolio marks become biased when liquid assets update faster than illiquid ones. Boundary mistakes can duplicate or omit events.

FORMULA INTUITION

For interval $[b_k, b_{(k+1)})$, assign each eligible event exactly once under left-closed, right-open rules. An as-of match requires $\text{source_time} \leq \text{decision_time}$ and $\text{age} \leq \text{tolerance}$.

SIMPLE MARKET EXAMPLE

A futures price at 10:00:00 and an ETF price last updated at 09:59:20 form a nominally synchronized row but represent different market states.

PYTHON / SYSTEM IMPLEMENTATION

Centralize bin closure, label, calendar, late-event, and empty-interval policies. Preserve source timestamp, matched timestamp, age, and finalization status after alignment.

CONFUSIONS TO AVOID

Upsampling adds rows, not information. Nearest-neighbor matching can select a future record unless direction is constrained.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The canonical time-grid service is shared by features, labels, validation, and execution. Quality gates reject rows with incompatible ages or provisional intervals.

RELATED CONCEPTS AND QUICK RECALL

Related: as-of join, resampling, watermark, Epps effect. Recall task: Define a safe synchronization rule for two assets with different update rates.

Cross-sectional and multivariate anomalies

BEGINNER

ONE-SENTENCE MEANING

Cross-sectional checks compare an instrument with peers at the same cutoff, while multivariate checks test whether combinations of fields are jointly plausible. They find semantic defects that univariate ranges miss.

MEMORY JOGGER

A value can be normal alone but impossible in combination. Peer context is powerful only when groups, currencies, and corporate states are comparable.

WHY IT MATTERS IN QUANT RESEARCH

Wrong multipliers, stale constituents, and broken joins often produce internally smooth series that diverge from peers. Joint checks catch these failures before factor ranking or covariance estimation.

FORMULA INTUITION

Robust cross-sectional score can use median and MAD by peer group; multivariate distance uses $d^2 = (x - \mu)' \Sigma^{-1} (x - \mu)$ with robust or regularized estimates.

SIMPLE MARKET EXAMPLE

Price and shares each fall within broad ranges, but their product makes market capitalization 100 times larger than any peer. The inconsistency points to a unit or identity error.

PYTHON / SYSTEM IMPLEMENTATION

Define point-in-time peer groups, normalize units, require coverage, and retain contributing fields for explainability. Use anomaly models as ranking evidence, not autonomous deletion.

CONFUSIONS TO AVOID

Contaminated peers and regime concentration can normalize a widespread defect. Classical covariance is unstable in high dimensions and can exaggerate distance.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The quality service combines hard constraints with peer and joint scores. Review dashboards expose which dimensions drove each alert and whether independent sources agree.

RELATED CONCEPTS AND QUICK RECALL

Related: peer group, Mahalanobis distance, robust covariance, entity resolution. Recall task: Construct a two-field inconsistency that would pass separate range checks but fail a joint economic identity.

Data-quality dimensions and scorecards

BEGINNER

ONE-SENTENCE MEANING

A quality scorecard separates completeness, validity, uniqueness, consistency, timeliness, coverage, and lineage instead of compressing them into one vague health number. Each dimension needs a denominator tied to expected data.

MEMORY JOGGER

A green average can hide a critical red component. Keep dimensions visible and connect each threshold to a consumer action.

WHY IT MATTERS IN QUANT RESEARCH

A feed can be complete but stale, timely but duplicated, or valid in schema while economically inconsistent. Multidimensional monitoring reveals which decisions remain safe.

FORMULA INTUITION

$\text{Rate} = \text{failing_expected_units} / \text{total_expected_units}$ for a declared grain and window. Aggregate with materiality weights only after preserving component values and uncertainty.

SIMPLE MARKET EXAMPLE

Ninety-nine percent row completeness is poor if the missing one percent contains the most liquid names during the open. Coverage by notional and consumer can be more informative than row count.

PYTHON / SYSTEM IMPLEMENTATION

Publish scorecards by source, field, instrument class, session, and policy. Include expected counts, observed counts, reason codes, trends, and affected consumers.

CONFUSIONS TO AVOID

A single quality score invites gaming and threshold confusion. Comparing rates across feeds without consistent expected denominators is meaningless.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Control planes translate dimension-specific states into warn, degrade, stop, or fallback actions. Owners and escalation targets appear beside each metric.

RELATED CONCEPTS AND QUICK RECALL

Related: completeness, validity, timeliness, coverage. Recall task: Design a scorecard that can show a feed is complete yet unusably stale.

Rule hierarchy and reason codes

BEGINNER

ONE-SENTENCE MEANING

A rule hierarchy orders structural invariants, semantic constraints, contextual diagnostics, and statistical anomaly evidence. Reason codes preserve which checks fired and which policy action followed.

MEMORY JOGGER

Hard facts should outrank soft surprise. A violated primary key is not equivalent to a large robust z-score.

WHY IT MATTERS IN QUANT RESEARCH

Prioritized rules prevent statistical heuristics from overriding exchange rules or corroborated market events. Explainable outcomes also make review, metrics, and policy changes safer.

FORMULA INTUITION

Example precedence: schema failure -> identity failure -> impossible economic state -> contextual warning -> statistical anomaly. Multiple reasons should remain attached even when one controls the action.

SIMPLE MARKET EXAMPLE

A negative price violates a hard domain rule, while a 12-sigma positive return is only suspicious until corroborated. The first can be quarantined automatically; the second needs context.

PYTHON / SYSTEM IMPLEMENTATION

Implement rules as versioned pure functions returning status, severity, reason, evidence, and suggested action. Test precedence and preserve all fired rules.

CONFUSIONS TO AVOID

One boolean clean flag cannot explain why a row disappeared. Letting later rules overwrite earlier outcomes destroys audit evidence and unstable policy behavior.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The policy engine maps rule outcomes to consumer-specific eligibility. Dashboards aggregate reason codes so engineering addresses causes rather than only symptoms.

RELATED CONCEPTS AND QUICK RECALL

Related: invariant, severity, policy engine, explainability. Recall task: Order five checks from structural certainty to statistical suspicion and justify the precedence.

Cross-source reconciliation

BEGINNER

ONE-SENTENCE MEANING

Reconciliation compares independent sources, aggregates, or accounting identities to detect omissions, semantic drift, and processing errors. Disagreement is evidence to investigate, not proof that either side is correct.

MEMORY JOGGER

Trust increases when separately constructed paths agree within understood tolerances. Shared upstream errors do not count as independent confirmation.

WHY IT MATTERS IN QUANT RESEARCH

Trades can reconcile to bars, positions to transactions, and internal prices to vendor benchmarks. These controls catch defects that row-level validators miss.

FORMULA INTUITION

$\text{residual} = \text{internal_measure} - \text{reference_measure}$; tolerance should reflect rounding, eligibility, latency, and source coverage. Track both absolute and economically scaled residuals.

SIMPLE MARKET EXAMPLE

Summed eligible trade size should match bar volume under the same policy. A difference may reveal lost events, duplicate delivery, or mismatched sale-condition filters.

PYTHON / SYSTEM IMPLEMENTATION

Document independence, transformation alignment, tolerance, ownership, and escalation. Drill residuals from portfolio to instrument to event with lineage.

CONFUSIONS TO AVOID

Two tables produced from the same corrupted normalized feed are not independent reconciliation. Forcing equality by overwriting one side removes the diagnostic.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Daily controls reconcile raw-to-curated counts, events-to-bars, actions-to-returns, and snapshots-to-consumers. Material breaks block publication or activate degraded mode.

RELATED CONCEPTS AND QUICK RECALL

Related: control total, tolerance, independent source, residual. Recall task: Propose one reconciliation for trades-to-bars and identify why two plausible systems might legitimately differ.

Coverage and distribution drift

BEGINNER

ONE-SENTENCE MEANING

Coverage drift changes which instruments, venues, times, or fields are observed, while distribution drift changes the values within that coverage. Monitoring both prevents composition changes from masquerading as market behavior.

MEMORY JOGGER

A feature mean can move because the world changed or because the dataset stopped seeing part of the world. Measure the denominator before interpreting the statistic.

WHY IT MATTERS IN QUANT RESEARCH

Universe loss, vendor policy changes, and new listings alter factor ranks and model inputs even when individual records pass all rules. Drift controls protect research comparability and live parity.

FORMULA INTUITION

Track coverage share by count, notional, venue, and expected updates; compare distributions with quantiles, PSI, Wasserstein distance, or task-specific metrics. Segment by regime and instrument class.

SIMPLE MARKET EXAMPLE

Average spread rises after a low-spread venue disappears from the feed. Without venue-mix monitoring, the change looks like a real liquidity shock.

PYTHON / SYSTEM IMPLEMENTATION

Maintain baseline windows, current windows, confidence bands, and composition tables. Link alerts to exact source, schema, policy, and affected feature versions.

CONFUSIONS TO AVOID

Drift is not automatically bad: crises and structural market changes are real. Automatically retraining on corrupted drift can teach the model a feed failure.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Monitoring separates data incident, market regime, and planned methodology change. The control plane can freeze models, switch sources, or lower limits while evidence is reviewed.

RELATED CONCEPTS AND QUICK RECALL

Related: population stability, coverage mix, regime drift, model monitoring. Recall task: Show how losing one venue can change a spread distribution without any true change in the remaining quotes.

Golden fixtures and property tests

BEGINNER

ONE-SENTENCE MEANING

Golden fixtures are small, hand-verified datasets with expected outputs, while property tests assert invariants across many generated cases. Together they test both known edge cases and broad rule behavior.

MEMORY JOGGER

Cleaning code needs examples of ugly reality, not only happy-path rows. A compact adversarial fixture can protect more than a large unreviewed sample.

WHY IT MATTERS IN QUANT RESEARCH

Boundary timestamps, duplicate retries, split events, zero MAD, stale quotes, and session resets frequently regress during refactors. Automated tests catch semantic changes before production data does.

FORMULA INTUITION

Properties include idempotence, deterministic output, key uniqueness, conservation under allowed transforms, no future timestamps, and stable partition recombination. Exact expectations cover policy-specific cases.

SIMPLE MARKET EXAMPLE

A fixture contains a trade at the bar boundary, one replay, one correction, and an early close. The expected bars prove interval closure, deduplication, correction linkage, and calendar handling together.

PYTHON / SYSTEM IMPLEMENTATION

Version fixtures with contracts, include counterexamples, and run unit, integration, replay, and batch-stream parity suites. Review golden updates as semantic changes.

CONFUSIONS TO AVOID

Snapshot tests can bless wrong output if reviewers do not understand the change. Random tests without reproducible seeds or economic constraints create noisy, weak evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

CI blocks schema or cleaning changes that violate invariants. Production canaries replay golden windows and compare hashes before full rollout.

RELATED CONCEPTS AND QUICK RECALL

Related: test fixture, property testing, regression test, parity. Recall task: Design one fixture that tests a duplicate, correction, and session boundary in the same event sequence.

Snapshot manifests and reproducibility

BEGINNER

ONE-SENTENCE MEANING

A snapshot manifest identifies the exact source partitions, schemas, policies, code, parameters, checksums, and quality reports that define a dataset. It turns a mutable collection of files into a reproducible research input.

MEMORY JOGGER

If an experiment cannot name its data snapshot, it cannot be reproduced. A path such as latest is operational convenience, not scientific identity.

WHY IT MATTERS IN QUANT RESEARCH

Model comparisons are invalid when inputs drift silently. Manifests make changes reviewable and allow precise rollback, impact analysis, and audit.

FORMULA INTUITION

`snapshot_id = hash(ordered file checksums, schema IDs, source cutoffs, transform versions, parameters)`. Logical equivalence rules must be explicit if byte order can vary.

SIMPLE MARKET EXAMPLE

Two researchers run the same notebook a week apart after a vendor revision. Binding each run to a manifest explains the performance difference without guessing.

PYTHON / SYSTEM IMPLEMENTATION

Publish atomically after validation; include row counts, min and max times, field statistics, quality status, parent snapshots, and owner. Make artifacts reference immutable IDs.

CONFUSIONS TO AVOID

A git commit does not capture external data, and a database timestamp alone does not identify changed rows. Mutable manifests defeat the purpose.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Training, backtesting, model registry, and live deployment all record dataset snapshot IDs. Rollback selects a previously verified manifest and compatible policy.

RELATED CONCEPTS AND QUICK RECALL

Related: manifest, content address, snapshot, model registry. Recall task: List the minimum manifest fields needed to reproduce a cleaned training dataset.

Incident response and safe publication

BEGINNER

ONE-SENTENCE MEANING

A data incident is a controlled response to evidence that published data may violate a consumer contract. Safe handling prioritizes protecting decisions, preserving evidence, scoping impact, rebuilding deterministically, and communicating status.

MEMORY JOGGER

Stop propagation before polishing the diagnosis. Never overwrite the evidence required to understand what happened.

WHY IT MATTERS IN QUANT RESEARCH

Bad data can affect orders, limits, features, models, and reports differently. A measured response prevents both uncontrolled use and unnecessary system-wide shutdown.

FORMULA INTUITION

Materiality combines affected rows, economic exposure, time range, consumers, and uncertainty. Recovery point and recovery time objectives should be defined for each data product.

SIMPLE MARKET EXAMPLE

A source begins scaling sizes by 100. Execution stops using depth immediately, risk falls back to a compatible source, and research snapshots are marked impacted pending rebuild.

PYTHON / SYSTEM IMPLEMENTATION

Declare incident, freeze affected publication, preserve raw payloads, identify first bad time, query lineage, activate fallback, repair under a new version, reconcile, and backfill with notice.

CONFUSIONS TO AVOID

Silently editing published files creates split-brain results. A fallback with different semantics can be more dangerous than temporary unavailability.

WHERE THIS FITS IN THE RESEARCH SYSTEM

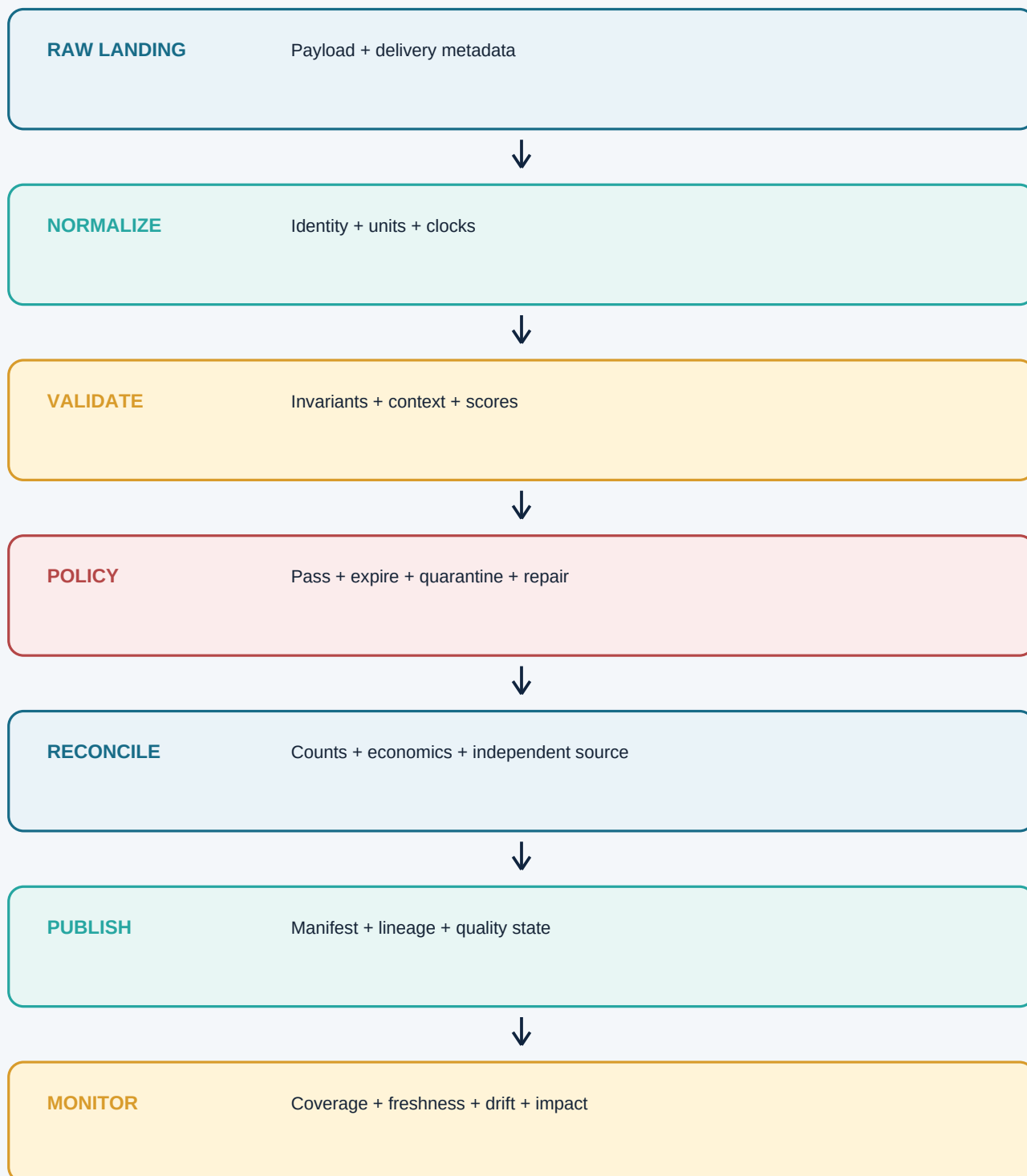
The data control plane links alerts, ownership, consumer actions, manifests, and recovery evidence. Post-incident review adds fixtures and prevention controls rather than only narrative.

RELATED CONCEPTS AND QUICK RECALL

Related: runbook, degraded mode, rollback, service-level objective. Recall task: Draft a first-15-minute response to a suspected 100x size scaling defect.

The evidence-preserving cleaning pipeline

BEGINNER



CONTROL PRINCIPLE

Raw evidence never flows backward from a repair. Every published value retains source identity, original value, rule outcomes, policy action, code version, and snapshot lineage.

Decision tree for a suspicious record

BEGINNER

Does schema, key, identity, unit, and clock parse correctly?

NO: quarantine structural failure

YES: test economic invariants

Does it violate an impossible domain or exchange rule?

YES: exclude; preserve evidence

NO: inspect context

Do independent sources or neighboring states confirm it?

YES: preserve likely real event

NO: score anomaly

Can policy derive a defensible repair without future data?

YES: publish repair plus trace

NO: quarantine or degrade

STOP RULE

If evidence is incomplete and the record can materially change a decision, do not invent certainty. Preserve it, restrict incompatible consumers, and escalate with full context.

Missingness patterns reveal different causes

BEGINNER

	Open	Midday	Close	Overnight	Revision
Liquid future	0	0	0	1	0
Small-cap equity	1	0	1	2	0
Halted equity	0	3	3	3	0
Venue B quotes	2	2	2	1	0
Fundamental field	1	1	1	1	2
FX conversion	0	0	2	2	0



0 Observed / expected



1 Sparse but plausible



2 Feed or join concern



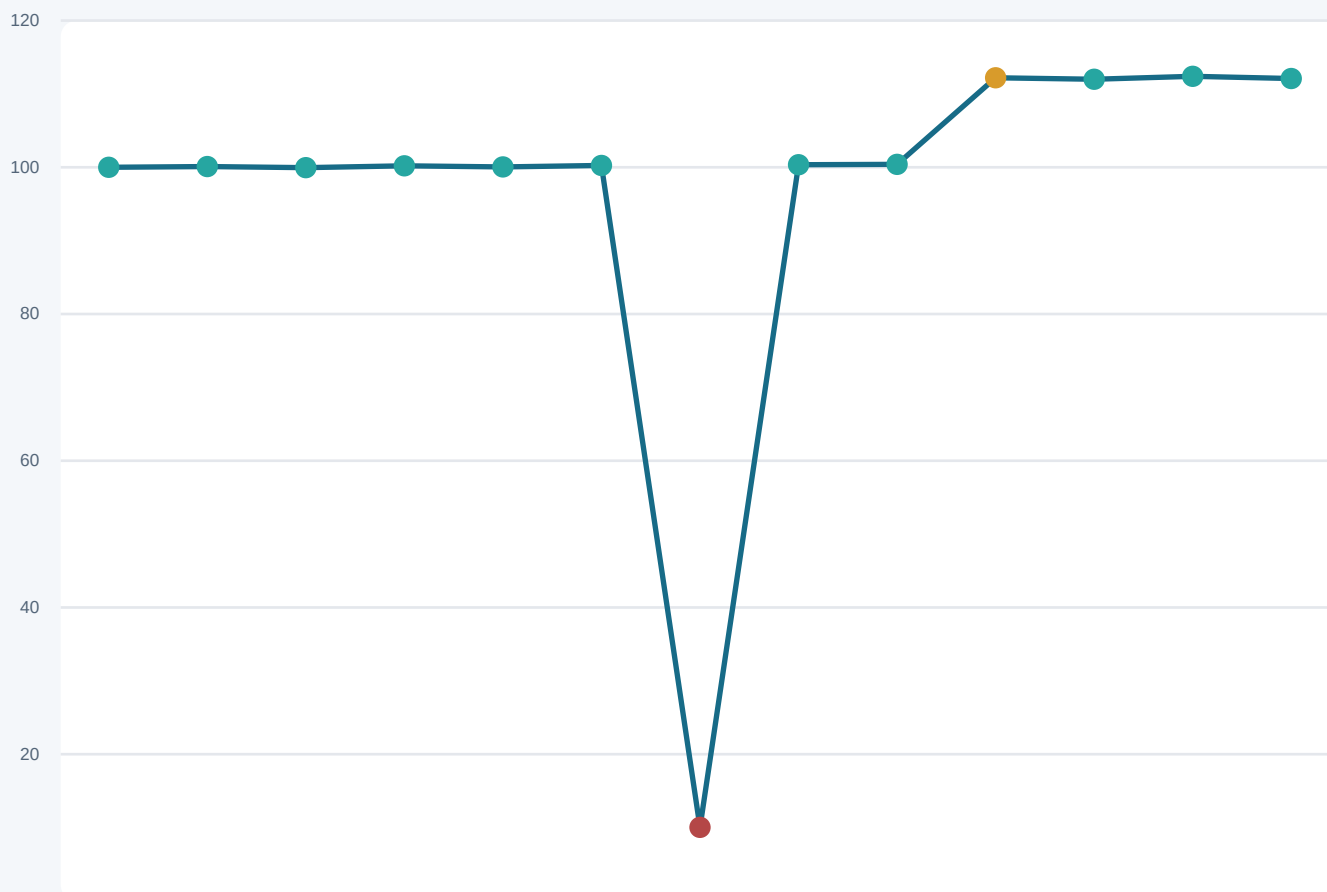
3 Structurally unavailable

INTERPRETATION GUIDE

Rows with the same null rate can have different mechanisms. Venue loss calls for source recovery, a halt is structurally unavailable, and a slow fundamental field needs an availability model rather than intraday filling. Measure against an expected grid and preserve the reason.

Bad tick diagnosis needs multiple signals

BEGINNER



POINT A - LIKELY DECIMAL ERROR

The isolated 10.03 print violates neighboring evidence and source scale while other venues remain near 100. Preserve the payload and quarantine it from bars until confirmation.

POINT B - LIKELY GENUINE REPRICING

The jump near 112 persists and receives confirming observations. Its anomaly score is large, but cross-source persistence supports a real event that must remain in tail analysis.

EVIDENCE STACK

Combine tick rules, bands, sequence health, quotes, independent sources, actions, and news state. Statistical magnitude alone cannot separate errors from legitimate jumps.

Repair methods by defect and consumer

BEGINNER

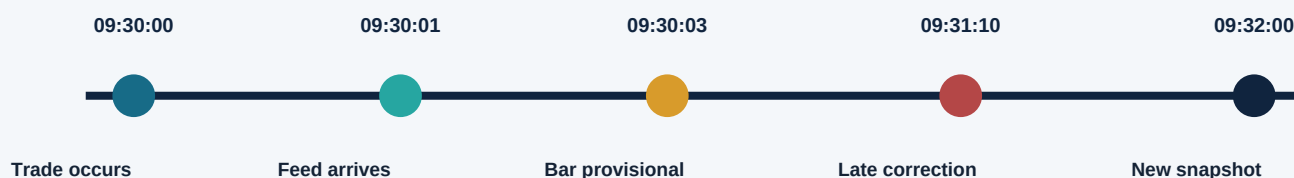
METHOD	DEFENSIBLE WHEN	MAIN RISK	REQUIRED TRACE
Exclude	Incompatible with consumer	Selection bias	Original + reason + scope
Expire	Last state is too old	Loss of coverage	Age + threshold + state
Forward fill	State persists; causal use	False freshness	Source time + max age
Interpolate	Offline bounded smooth field	Future leakage	Endpoints + method
Winsorize	Feature influence needs control	Destroyed tails	Bounds + cap indicator
Rebuild	Derived output is reproducible	Policy mismatch	Events + policy + hash
Cross-source	Authoritative replacement exists	Semantic mismatch	Source rank + tolerance
Quarantine	Material evidence is uncertain	Delayed availability	Evidence + owner + SLA

SELECTION RULE

Prefer the least assumptive action that preserves causal meaning. If a method creates a value, attach synthetic status, original evidence, uncertainty, boundaries, and the consumer contract that permits its use.

Point-in-time validation timeline

BEGINNER



DECISION AT 09:30:00.5

The trade has occurred but has not arrived. A realistic system cannot use it, even if end-of-day data later sorts it into the correct event-time location.

DECISION AT 09:30:02

The trade is eligible and the current bar is provisional. Consumers must know whether provisional values are permitted and what revision risk remains.

DECISION AFTER CORRECTION

The correction creates a new active version; earlier snapshots remain unchanged. Rebuilding all history with the correction would erase the knowledge path.

VALIDATION TEST

Add or modify records strictly after an earlier cutoff and recompute. Features and quality decisions before that cutoff must remain identical.

CAUSAL RULE

Use availability time for what the system knew, event time for market order, and effective intervals for economic applicability. Never collapse these clocks into one timestamp.

Quality scorecard by source and dimension

BEGINNER

	Complete	Valid	Unique	Timely	Coverage	Lineage
Trades A	PASS	PASS	PASS	PASS	PASS	PASS
Quotes A	PASS	PASS	PASS	WARN	PASS	PASS
Quotes B	WARN	PASS	PASS	STOP	STOP	PASS
Actions	PASS	WARN	PASS	PASS	PASS	WARN
Fundamentals	WARN	PASS	PASS	WARN	WARN	PASS
FX rates	PASS	PASS	PASS	STOP	WARN	PASS

WHY DIMENSIONS STAY SEPARATE

Quotes B passes validity and uniqueness but fails timeliness and coverage. A single average might appear acceptable, yet execution should stop using it while bounded offline research may continue.

CONSUMER-AWARE ACTION

Each cell maps to warn, degrade, stop, or fallback by consumer. A stale quote is critical for routing, material for intraday features, and irrelevant to some audits.

DENOMINATOR DISCIPLINE

Completeness uses expected records, timeliness uses declared latency, and coverage uses eligible instruments or notional. Publish denominators beside every percentage.

Failure modes and first diagnostic actions

BEGINNER

SYMPTOM	LIKELY CAUSE	CHECK	ACTION
Volume doubles after retry	Duplicate delivery	Event IDs and offsets	Replay idempotently
One-day -50% return	Unapplied split	Action factor and dates	Rebuild adjusted view
Spread turns negative	Stale or async quote	Leg ages and sequences	Reconstruct state
Smooth outage period	Forward fill hid gap	Observed flag and age	Expire synthetic state
Feature rows multiply	Join cardinality	Keys and validity ranges	Fix grain before join
Backtest changes weekly	Mutable vendor history	Snapshot diffs and vintages	Bind immutable manifest
Liquidity jumps 100x	Unit or multiplier drift	Contract metadata	Version normalization
Open differs by run	Unstable event order	Sequence and tie-break	Enforce total order
Only latest symbols survive	Identity backfill	Effective mappings	Restore point-in-time IDs

TRIAGE ORDER

Protect consumers, freeze publication, preserve evidence, scope the first bad time, verify identity, time, and units, rebuild deterministically, reconcile, and republish under a new manifest.

Formula and validation sheet

BEGINNER

ROBUST SCALE

$MAD = \text{median}(|x_i - \text{median}(x)|)$; normal-consistent scale is about $1.4826 * MAD$. Use field-specific zero-scale and minimum-sample rules.

CAUSAL ROLLING SCORE

$z_t = (x_t - \text{center}_{(t-1, W)}) / \text{scale}_{(t-1, W)}$. The reference window ends before the scored observation and resets at declared boundaries.

OBSERVATION AGE

$\text{age}_t = \text{decision_time} - \text{last_update_time}$. Interpret age using session, instrument, source, and expected update intensity.

BAR INVARIANTS

$\text{low} \leq \min(\text{open}, \text{close}) \leq \max(\text{open}, \text{close}) \leq \text{high}$ and $\text{volume} \geq 0$. Event reconciliation is still needed to prove the path.

AS-OF ELIGIBILITY

$\text{available_time} \leq \text{cutoff}$, $\text{effective_from} \leq \text{cutoff} < \text{effective_to}$, active revision, correct identity, and age within tolerance.

LINEAR INTERPOLATION

$x(t) = x_a + (x_b - x_a)(t - t_a) / (t_b - t_a)$. Endpoint b is future information for decisions before t_b .

MULTIVARIATE DISTANCE

$d^2 = (x - \mu)' \Sigma^{-1} (x - \mu)$. Use robust or regularized estimates and retain contributing dimensions.

RECONCILIATION RESIDUAL

$\text{residual} = \text{internal-reference}$. Scale tolerance by rounding, eligibility, latency, coverage, and economic materiality.

SNAPSHOT IDENTITY

Hash ordered checksums, schema IDs, source cutoffs, code, policy, parameters, and quality report.

Worked example: bad tick to repaired snapshot

BEGINNER

1. DETECT

A one-minute bar shows a -89% close and a 40x volume spike. The monitor links the bar to one trade whose price is one decimal place below simultaneous quotes.

2. PRESERVE

The raw payload, channel, sequence, arrival time, and checksum remain immutable. The bar and affected features are marked provisional.

3. VALIDATE STRUCTURE

Schema and identity pass, but the trade violates venue price bands and source tick grid. Sequence continuity shows no missing messages.

4. SEEK CONTEXT

Other venues, consolidated quotes, and later prints remain near the prior level. No action, halt, auction transition, or broad repricing is present.

5. DECIDE

Policy classifies the print as a likely decimal error and excludes it from regular-session bars. The original remains in the all-events audit view.

6. REBUILD

The bar is regenerated from eligible events; OHLC, volume, and count reconcile to source rows. Downstream rolling features update only affected windows.

7. PUBLISH

A new manifest records source IDs, old and new bar values, reason, policy, code hash, lineage, and quality approval.

8. IMPACT ANALYSIS

Catalog lineage identifies experiments, reports, and models that consumed the defective snapshot. Live consumers switch only after parity checks.

9. PREVENT RECURRENCE

A golden fixture adds the decimal error plus a genuine confirmed gap, proving the rule protects legitimate tail behavior.

Dataset promotion checklist

BEGINNER

CONTRACT

Row grain, key, field meaning, unit, clock, calendar, valid states, expectedness, correction policy, and consumers are declared.

EVIDENCE

Raw payloads, delivery metadata, source identity, checksums, and original values are immutable and queryable.

RULES

Structural, semantic, contextual, statistical, and reconciliation checks return versioned reasons with defined precedence.

REPAIRS

Exclusion, expiry, fill, cap, replacement, and rebuild policies preserve originals, boundaries, uncertainty, and causal availability.

TESTS

Fixtures, properties, replay, idempotence, batch-stream parity, future perturbation, and reconciliation pass.

PUBLICATION

Manifest, lineage, quality report, limitations, owner, source cutoffs, schemas, code, and checksums are atomically bound.

OPERATIONS

Metrics have denominators, thresholds map to actions, fallbacks are compatible, and runbooks have recovery objectives.

PROMOTION GATE

No unresolved material defect, future dependency, unexplained revision, or unowned alert remains.

Final active recall and system index

BEGINNER

START WITH MEANING

Cleaning is use-case-specific semantic control, not universal smoothing. Define grain, identity, clocks, units, and consumer first.

PRESERVE EVIDENCE

Immutable raw data, correction links, vintages, and row lineage make every repair reversible and auditable.

CLASSIFY ABSENCE

Expected gaps, outages, invalid records, censored values, and not-yet-available facts require separate reasons and actions.

RESPECT REAL TAILS

Outlier scores measure surprise, not truth. Corroborate with rules, neighboring state, independent sources, actions, and regime.

CREATE VALUES CAREFULLY

Fills, interpolation, caps, and replacements are derived data. Preserve status, source endpoints, boundaries, age, and uncertainty.

VALIDATE SYSTEMS

Keys, identities, unit conservation, reconciliation, snapshot hashes, fixtures, and causal perturbation tests protect the pipeline.

PUBLISH IMMUTABLY

Bind data, code, policy, schema, quality, lineage, and compatible consumers in a manifest rather than latest.

OPERATE EXPLICITLY

Monitor completeness, validity, uniqueness, timeliness, coverage, drift, and impact. Map failures to safe actions.

FINAL RECALL

For any cleaned value, explain original evidence, fired rules, policy action, availability, uncertainty, snapshot, and consumer.